

Preliminary MH370 hydroacoustic work

Material for discussion with Dr Usama Kadri — 24 August 2026

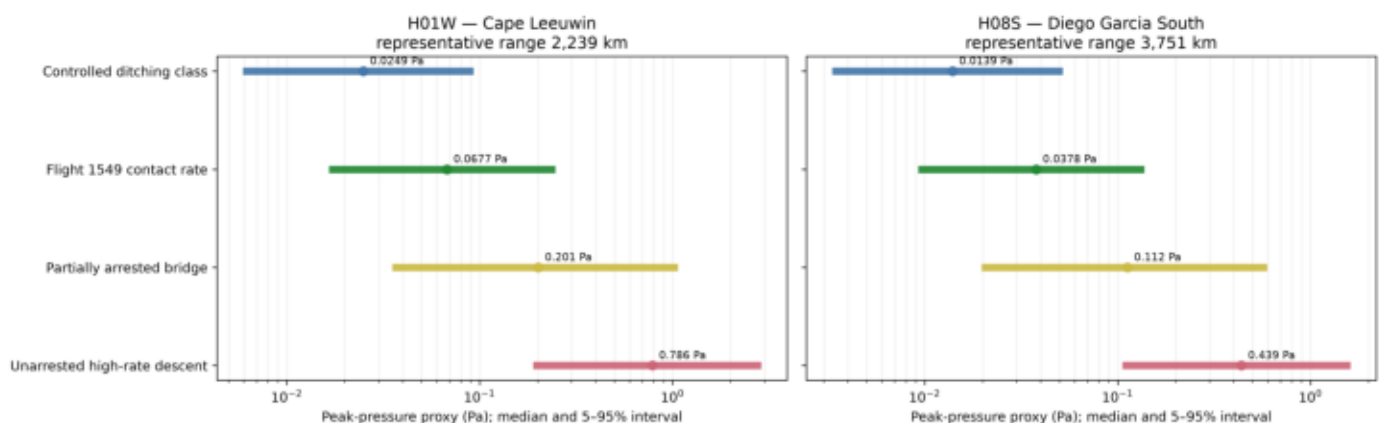
This work asks two separate questions: what pressure scale might different Boeing 777 water-impact conditions couple into the ocean, and whether any feature visible in Kadri's published H01W/H08S plots can presently be associated with such an impact.

The simplified impact-to-SOFAR sensitivity experiment uses one million deterministic draws for each of four explicitly separate vertical-contact-speed families and two source-to-SOFAR coupling priors. It uses 174,369 kg—the actual 9M-MRO zero-fuel weight on the final loadsheet—as a fuel-exhaustion impact-mass surrogate, not a measured mass at 00:19 UTC. The F-35 example only centres one coupling prior at order 10^{-4} ; the log-uniform control spans 10^{-8} to 10^{-2} . Under the centred prior, median peak-pressure proxies range from 0.025 to 0.786 Pa at H01W and 0.014 to 0.439 Pa at H08S. These are not detection probabilities and do not identify a plotted transient.

The present candidate assessment is deliberately conservative. Kadri's Figure 9 rectangle 1/Table 1 caption identify an approximately 00:52 UTC, 57-degree signal; rectangle 2/Table 1 identify the preferred 00:54:30 UTC, 306.18-degree candidate. The p. 9 prose appears to conflate their time and bearing. Neither the 00:49:58 nor 00:53:31 Table 1 time exceeds our publication-panel q99 screen within ± 2 , ± 5 or ± 10 seconds. The 01:03:13.15 H08S screening time is elevated in the plotted trace but lies within a 9.9279-second periodic airgun train. Leave-one-shot-out subtraction leaves its residual between the 80.5th and 99.2nd percentile depending on defensible method choice; the control-selected result is 94.1st percentile.

At ± 100 and ± 150 ms, masks centred on the blind periodic prediction leave the screened time, whereas masks centred on observed shot peaks remove it. Both tests operate below the publication line's indicative ± 0.51 -second timing thickness, so the plotted waveform cannot adjudicate between them.

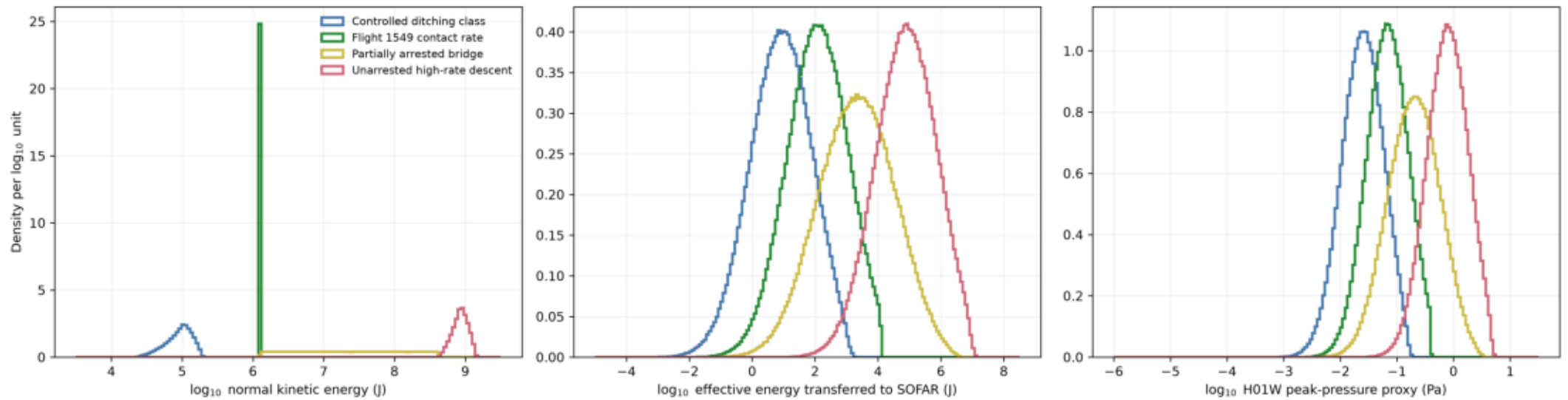
Current conclusion: hydroacoustics should enter the integrated estimator only through an explicitly labelled candidate-association hypothesis not as an unconditional observable or non-detection likelihood. Raw triad channels, exact processing metadata, airgun timing/source information, and a declared detector/noise model would be required to change that status.



All results are preliminary and conditional; individual vector PDF figures remain the publication masters.

Impact-to-SOFAR Monte Carlo sensitivity

Impact-family transfer sensitivity under the F-35-centred coupling prior
Family-conditional Monte Carlo; not an empirical event posterior



Families and coupling priors are explicit model alternatives; the curves are not an impact or detection posterior.

Candidate bearings, arrival-difference bands and integrated spatial PDF

Kadri source-signal timing, candidate bearings and two-station timing controls

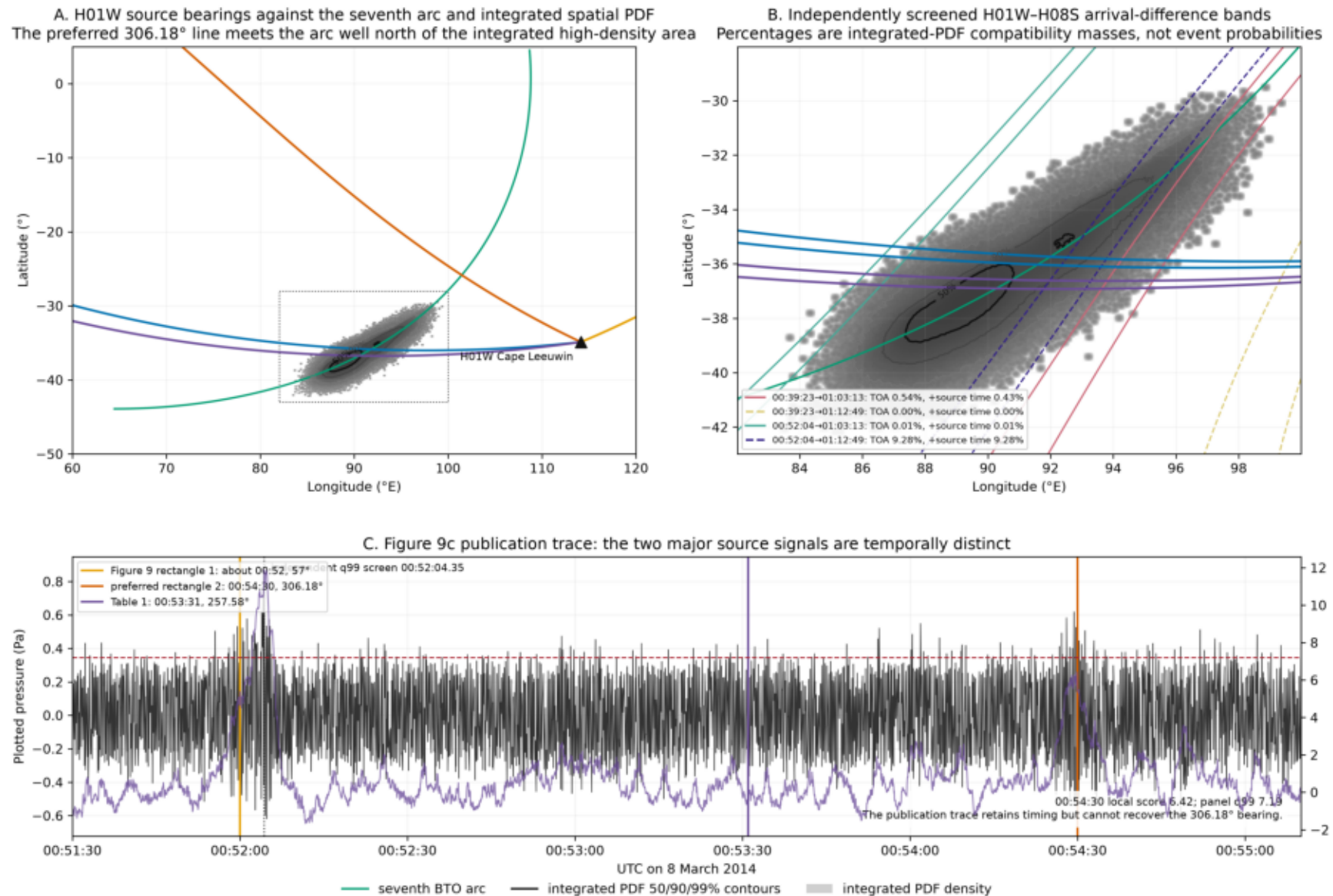
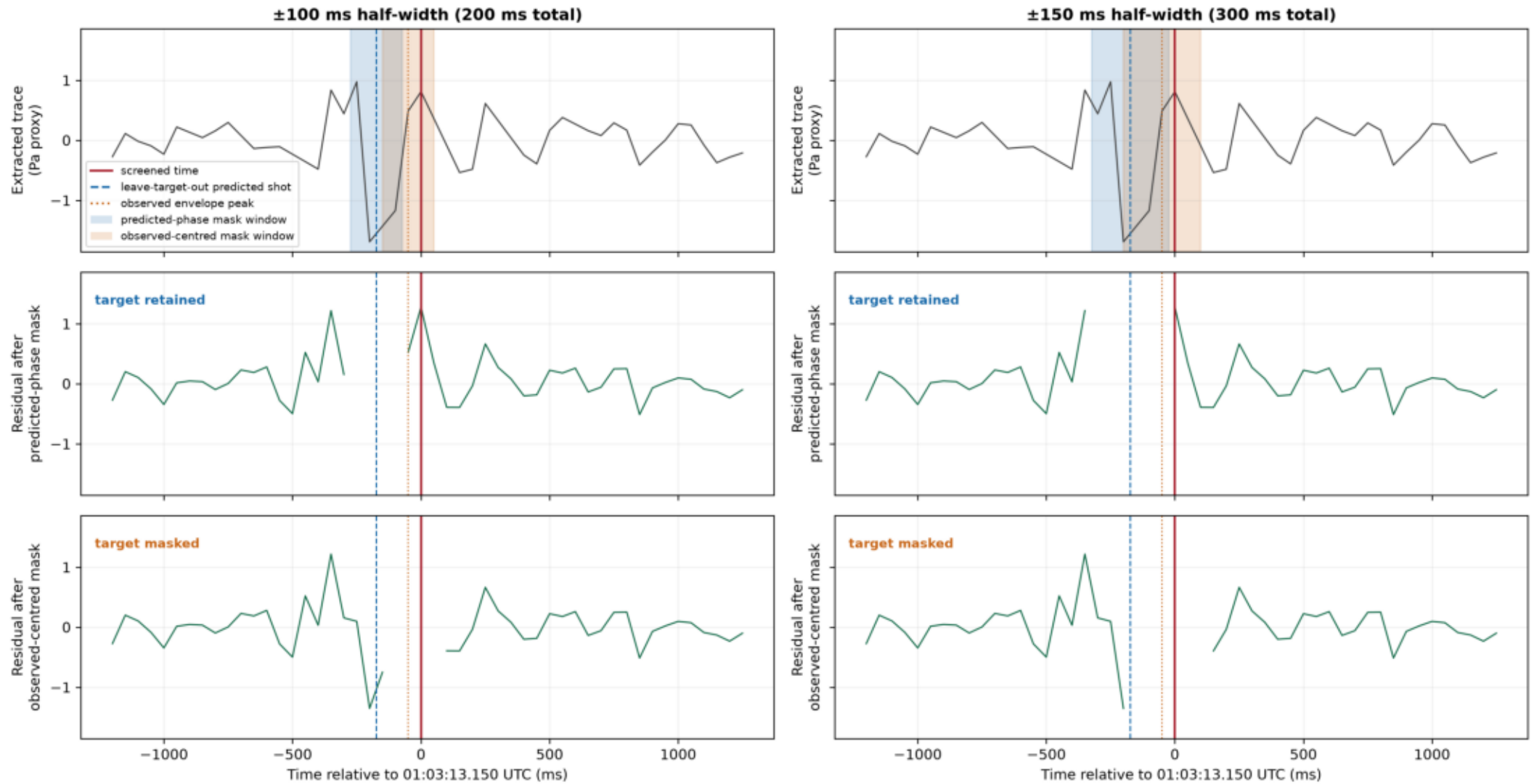


Figure 9 rectangle 1 (~00:52/57°) and preferred rectangle 2 (00:54:30/306.18°) are distinct; the p. 9 prose appears to conflate them. Arrival-difference bands cancel impact time; the second percentage also enforces impact during 00:19:37-00:49:37.

Source signals and independently screened timing controls are separate categories; compatibility mass is not event probability.

Diego Garcia $\pm 100/\pm 150$ ms mask sensitivity

H08S sub-second periodic-mask sensitivity



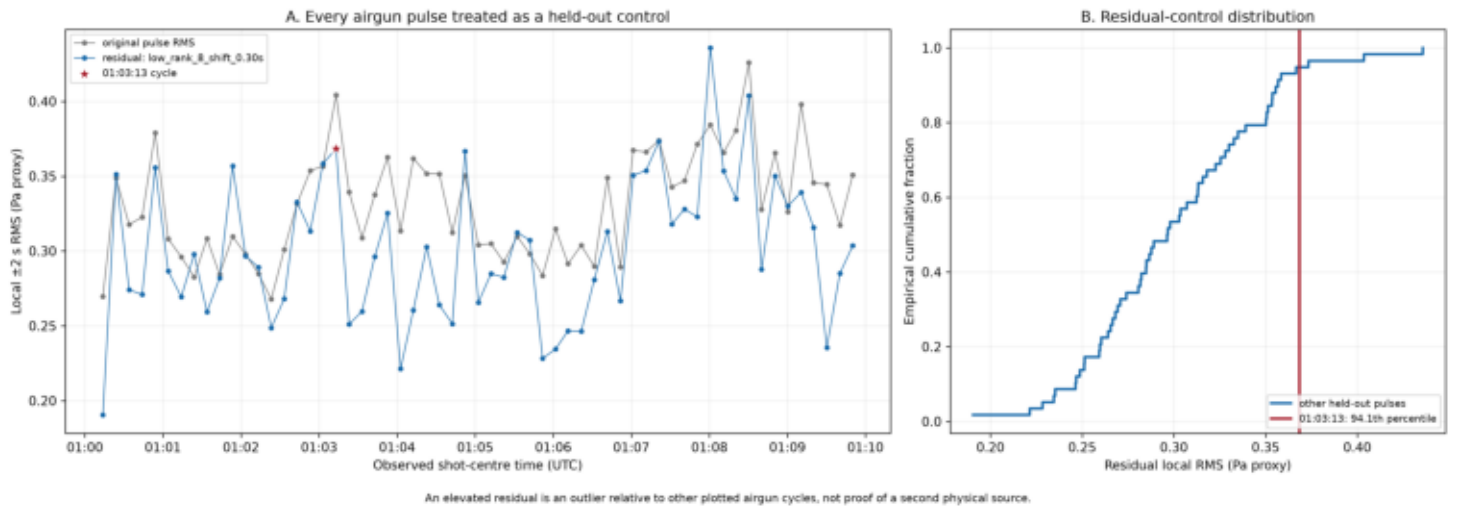
The screened time is +174 ms from the target-excluded periodic prediction and +50 ms from the observed peak. Both widths are below the publication line's indicative ± 0.51 s timing thickness; raw channels are required.

The apparent result changes with mask centring, and both widths are below publication-rendering timing precision.

Diego Garcia periodic-interference controls

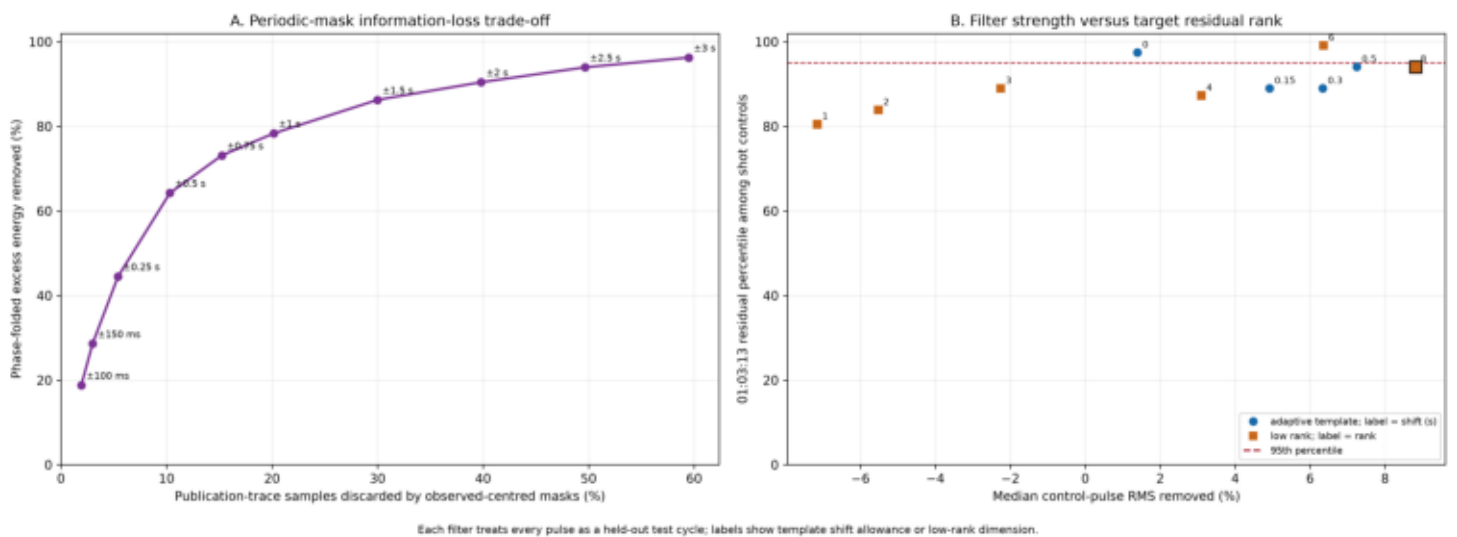
Held-out airgun-cycle controls

Leave-one-shot-out test of the 01:03:13 H08S feature



Filter and information-loss sensitivity

H08S periodic-airgun filter and mask sensitivity



An elevated residual is not a source classifier; raw triad channels are required for a defensible detector.